TCP Question — extra semester test 1

Assume processes A and B are communicating using TCP. For the sake of this question assume that all actions are atomic; when a message is sent,it is received essentially immediately at the other end. Stated differently, two events will never overlap, with one exception — see events 3 and 4 below.

Assume A has just sent a message to B with SEQ=100, LEN=500,ACK=1000 and WIN=2000. Immediately after that B sent a message to A with LEN=300 and WIN=1000. Then the following events happen in sequence outlined below. However, events 3 and 4 happen at exactly the same moment and these messages both arrive at their destinations a split second later.

1. A sends a message to B with LEN=200.
2. B consumes 500 bytes from its buffer.
3. A sends a message to B with LEN=600.
4. B sends a message to A with LEN=200.
5. A consumes 1000 bytes from its buffer.
6. A sends a message to B with LEN=300

Below events 1, 3, 4 and 6 will be referred to as messages 1, 3, 4 and 6, respectively.

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Things to note in this memo:

- The two lines at each end contain the values before and after the event.
- The 'before' values are simply copied from the 'after' values at the previous event.
- Sending a message causes two changes at each communicating party:
  1. At the sender LEN is added to SEQ and subtracted from the window side at the other end.
  2. At the recipient LEN is added to ACK and subtracted from the own window.
  3. Whatever a party learns from a message about the other side takes precedence over its own records.[1]
- Note the challenges to manage the information — I have to use a tiny font / large sheet of paper.

[1]Think about the correctness of this claim where messages 3 and 4 are sent simultaneously. In fact, it cannot be correct... It doesn't affect the test answers, though.

- The values that accompany messages are obviously those that are in the 'before' line of the sender.

SEQ	ACK	Win	Win@B
100	1000	2000	?
100+500			?-500

⟶A sends to B: SEQ=100, LEN=500,ACK=1000 and WIN=2000

SEQ	ACK	Win	Win@A
1000	100	?	?
	100+500	?-500	2000

SEQ	ACK	Win	Win@B
600	1000	2000	?
	1000+300	2000-300	1000

⟵B sends 300 to A, Win=1000, LEN=300

SEQ	ACK	Win	Win@A
1000	600	1000	2000
1000+300			2000-300

SEQ	ACK	Win	Win@B
600	1300	1700	1000
600+200			1000-200

⟶Event 1: A sends a message to B with LEN=200.

SEQ	ACK	Win	Win@A
1300	600	1000	1700
	600+200	1000-200	1700

Event 2: B consumes 500 bytes from its buffer.

SEQ	ACK	Win	Win@A
1300	800	800	1700
		800+500	

SEQ	ACK	Win	Win@B
800	1300	1700	800
800+600			800-600

Events 3 and 4: A sends a message to B with LEN=600; B sends a message to A with LEN=200.

SEQ	ACK	Win	Win@A
1300	800	1300	1700
1300+200			1700-200

SEQ	ACK	Win	Win@B
1400	1500	1500	700

Events 3 and 4 complete. (Messages arrive.)

SEQ	ACK	Win	Win@A
1500	1400	700	1500

SEQ	ACK	Win	Win@B
1400	1500	1500	700
		1500+1000	

Event 5: A consumes 1000 bytes from its buffer.

SEQ	ACK	Win	Win@B
1400	1500	2500	700
1400+300			700-300

⟶Event 6: A sends a message to B with LEN=300

SEQ	ACK	Win	Win@A
1500	1400	700	1500
	+300	700-300	2500